Defect Analysis of PVC Pipe using Statistical Quality Control (SQC)

To develop a preventive quality monitoring framework using control charts.

Abstract:

his study presents a defect analysis of PVC pipe manufacturing using Statistical Quality Control (SQC) techniques. Defects such as cracks, melt distortion, and hollow formation can affect product quality and increase production costs. A simulated dataset consisting of 20 production batches with 200 pipes inspected per batch was analyzed to evaluate defect patterns and process stability. Pareto analysis was used to identify the most significant defect types, followed by Cause-and-Effect (Ishikawa) diagrams to examine potential root causes of major defects. An NP-control chart was constructed to assess statistical control of the process. The results indicate that crack defects represent the largest proportion of total defects. Control chart analysis shows that the process remains within statistical control limits, though variation exists across batches. Based on the findings, a preventive quality monitoring framework using control charts is proposed to ensure early detection of deviations and continuous improvement in PVC pipe production.

Introduction:

Quality plays a critical role in modern manufacturing systems, particularly in industries where product reliability directly impacts safety and customer satisfaction. PVC (Polyvinyl Chloride) pipes are widely used in water supply systems, irrigation networks, drainage systems, and construction applications. Due to their extensive usage, maintaining consistent quality standards in PVC pipe production is essential.

During the manufacturing process, defects such as cracks, melt distortion, and hollow formation may occur due to improper temperature control, machine settings, raw material variation, or operational inconsistencies. These defects not only reduce product strength and durability but also increase rejection rates and production costs. Therefore, systematic quality monitoring becomes necessary to ensure process stability and minimize variation.

Statistical Quality Control (SQC) provides scientific tools to monitor, analyze, and control production processes using statistical methods. Control charts, Pareto analysis, and defect trend evaluation help manufacturers identify dominant quality issues and determine whether a process is operating within acceptable limits. In addition, cause-and-effect analysis supports systematic identification of potential root causes contributing to major defects. By distinguishing between random variation and assignable causes, SQC supports data-driven decision-making in production environments.

This study aims to analyze defect patterns in PVC pipe manufacturing using SQC techniques and to develop a preventive quality monitoring framework. Through defect classification, control chart analysis, and economic evaluation, the research demonstrates how statistical tools can enhance process stability and support continuous quality improvement.

Problem Statement:

PVC pipe manufacturing processes often experience defects such as cracks, melt distortion, and hollow formation, which reduce product quality and increase rejection costs. Variations in machine settings, raw material conditions, and operational factors may contribute to inconsistent production outcomes. Without systematic statistical monitoring, it becomes difficult to distinguish between normal process variation and assignable causes of defects. Therefore, there is a need to analyze defect patterns and evaluate process stability using Statistical Quality Control techniques in order to support preventive quality monitoring and reduce defective output.

Objectives:

The primary objective of this study is to analyze defects in PVC pipe manufacturing using Statistical Quality Control techniques. The specific objectives are:

1. To identify the major types of defects occurring in PVC pipe production.
2. To analyze defect frequency and distribution using Pareto analysis.
3. To identify potential root causes of major defects using a Cause-and-Effect diagram.
4. To evaluate process stability using an NP-control chart.
5. To estimate the economic impact of defective production.
6. To develop a preventive quality monitoring framework using control charts for continuous process improvement.

Literature Review:

Quality control in manufacturing has been widely studied as a critical factor in improving productivity and reducing waste. Statistical Quality Control (SQC) techniques are commonly used to monitor process variation and identify assignable causes of defects. Control charts, introduced by Walter A. Shewhart, are fundamental tools for distinguishing between random variation and process abnormalities.

Several studies in manufacturing industries have applied control charts and Pareto analysis to identify dominant defect categories and assess process stability. Pareto analysis helps prioritize critical quality issues by focusing on the most significant contributors to defects, while control charts provide a graphical representation of process performance over time.

In PVC pipe production, defects such as cracks, surface irregularities, and material distortion are often linked to variations in temperature, extrusion speed, and raw material quality. Previous research has demonstrated that systematic statistical monitoring improves defect detection and supports data-driven decision-making. However, many studies primarily focus on defect identification rather than developing a structured preventive monitoring framework.

This study extends existing applications of SQC by integrating defect analysis with a preventive quality monitoring approach using control charts to support continuous process control.

In addition to control charts and Pareto analysis, Cause-and-Effect (Ishikawa) diagrams are widely used in quality management to explore potential root causes of defects. This tool complements statistical monitoring by providing a structured method to analyze production-related factors influencing defect generation.

Research Methodology:

1. Research Design

This study adopts a quantitative analytical approach to evaluate defects in PVC pipe manufacturing using Statistical Quality Control (SQC) techniques. The analysis focuses on identifying defect patterns, assessing process stability, and proposing a preventive monitoring framework.

2. Data Collection

Due to limited access to industrial production data within the available timeframe, a simulated dataset was generated to represent realistic PVC pipe manufacturing conditions. The dataset reflects typical defect rates observed in manufacturing environments.

The data consist of:

- 20 production batches
- 200 PVC pipes inspected per batch
- Three defect categories: Crack, Melt, and Hollow

The simulated data were structured to reflect normal production variability without extreme abnormalities.

3. Sampling Method

A fixed sample size approach was used, where 200 pipes were inspected from each production batch. This allows consistent comparison across batches and enables the use of attribute control charts for statistical analysis.

4. Statistical Tools Used

The following Statistical Quality Control tools were applied:

a) Pareto Analysis

Used to identify the most significant defect category contributing to total defective production.

b) Cause and Effect Diagram

Used to identify the possible causes responsible for major defects in PVC pipes. This tool helps in understanding the relationship between defects and various production factors such as machine settings, materials, methods, and manpower.

c) NP-Control Chart

An NP-control chart was constructed to monitor the number of defective pipes per batch. Since the sample size remains constant across all batches, the NP-chart is appropriate for evaluating process stability.

Control limits were calculated using standard statistical formulas to determine whether the process operates within acceptable variation.

d) Trend Analysis

A batch-wise defect trend graph was plotted to observe variations and identify any systematic increase or decrease in defect levels.

e) Cost Impact Analysis

An estimated cost per defective pipe was assumed to evaluate the economic impact of defects on overall production.

5. Data Analysis Procedure

The collected data were tabulated and analyzed using spreadsheet software. Control limits for the NP-chart were calculated using standard SQC formulas. Graphical representations were prepared to interpret process behaviour and support preventive monitoring recommendations.

This methodology ensures systematic evaluation of defect patterns and provides a structured framework for preventive quality control in PVC pipe manufacturing.

Data Presentation:

To evaluate defect patterns in PVC pipe manufacturing, a dataset consisting of 20 production batches was prepared. For each batch, 200 PVC pipes were inspected and categorized based on the type of defect observed. The defects were classified into three main categories:

- Crack
- Melt Distortion

- Hollow Formation

The batch-wise defect data are presented in Table 1.

Table 1: Batch-wise Defect Data (PVC Pipe Production)

Batch	Crack	Melt	Hollow	Total Defective
1	6	4	3	13
2	7	5	2	14
3	9	6	3	18
4	11	7	3	21
5	8	6	2	16
6	10	8	3	21
7	7	5	2	14
8	12	7	4	23
9	6	4	2	12
10	8	5	3	16
11	9	6	3	18
12	13	8	4	25
13	7	5	2	14
14	6	4	3	13
15	8	6	2	16
16	10	7	3	20
17	9	6	2	17
18	11	8	3	22
19	7	5	2	14
20	8	6	3	17
Total	172	118	54	344

Each batch consisted of 200 inspected pipes, resulting in a total inspection size of 4,000 pipes across all batches. The data presented above form the basis for subsequent statistical analysis using Pareto analysis, NP-control chart, and trend evaluation.

Data Analysis and Interpretation:

Statistical analysis was conducted to evaluate defect patterns and process stability in PVC pipe manufacturing. The purpose of this analysis is to identify the most critical defects, assess whether the production process is statistically stable, examine variation trends, and estimate the financial impact of defects. The following Statistical Quality Control tools were applied to achieve these objectives.

1. Pareto Analysis

Pareto analysis is used to identify the most significant factors contributing to a problem. It is based on the 80/20 principle, which states that a small number of causes often contribute to the majority of the problem. In quality control, Pareto charts help prioritize defect categories so that improvement efforts can focus on the most impactful issues.

A Pareto chart is a combination of:

- A bar graph representing frequency of defects
- A cumulative percentage line indicating overall contribution

This tool helps distinguish critical defects from less significant ones.

Total defects observed across 20 batches were:

- Crack = 172
- Melt = 118
- Hollow = 54

Total Defects = 344

Percentage contribution:

- Crack = 50%
- Melt = 34%
- Hollow = 16%

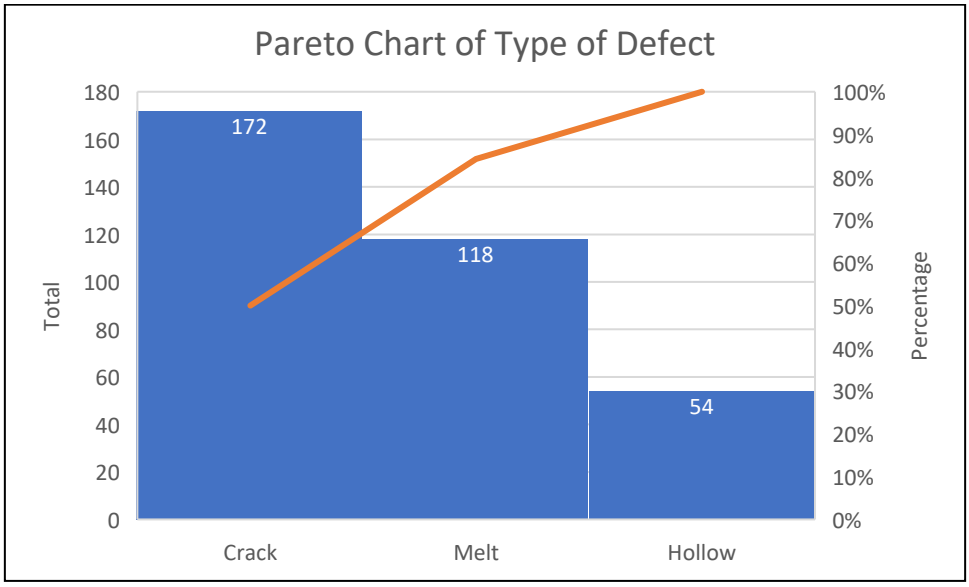

Figure 1: PVC pipe pareto chart

The Pareto principle suggests that a majority of production problems are typically caused by a limited number of defect categories. In this study, crack defects (50%) and melt defects (34%) together account for 84% of total PVC pipe defects. Since their cumulative contribution is close to 80%, these two defect types have the greatest impact on overall product quality. Therefore,

focusing corrective measures on crack and melt defects is expected to significantly reduce the overall defective rate in PVC pipe production.

2. Cause and Effect Analysis

A Cause-and-Effect Diagram, also known as the Ishikawa or Fishbone diagram, is a quality control tool used to systematically identify and analyze the possible causes of a specific problem. It helps in categorizing potential root causes under structured headings such as Machine, Material, Method, Manpower, Measurement, and Environment (6M framework). This tool supports problem-solving by visually organizing contributing factors and guiding corrective and preventive actions.

Based on Pareto analysis, crack (50%) and melt (34%) defects together account for approximately 84% of total PVC pipe defects. Therefore, separate cause-and-effect diagrams were developed to identify potential root causes for both defect types.

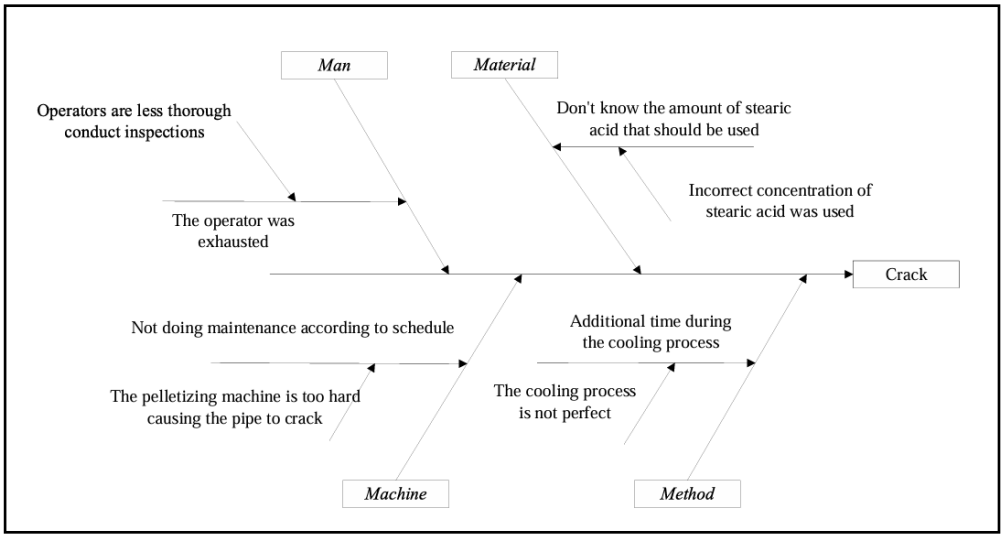

Figure 2: Cause and Effect diagram of crack defective

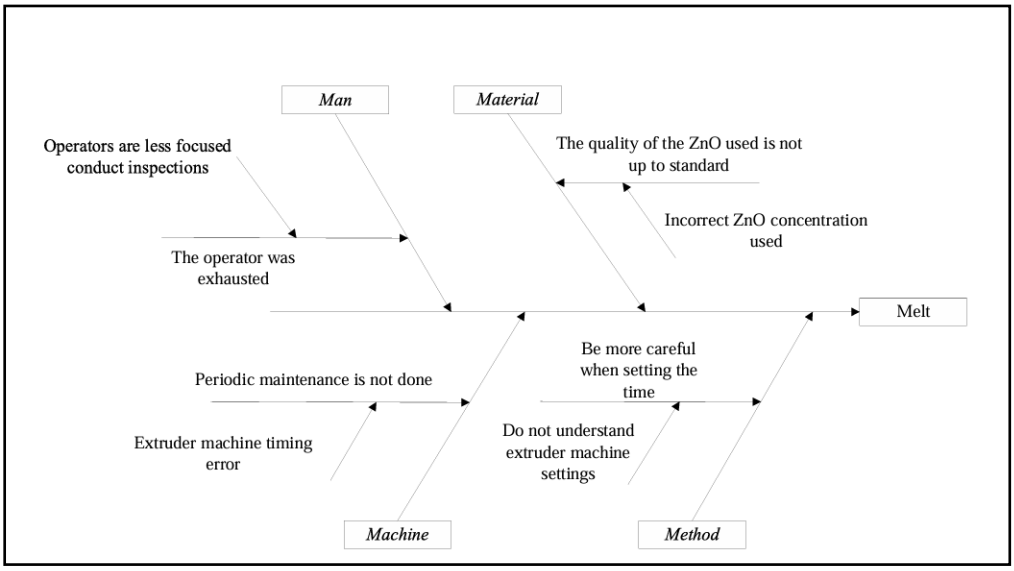

Figure 3: Cause and Effect diagram of Melting defective

Based on the picture above, it can be seen that the causes of disability can be caused by humans, machines, materials, and the methods used. So that improvements can be made to these four aspects.

3. NP-Control Chart Analysis

Control charts are used to determine whether a production process is statistically stable. They help distinguish between:

- Common cause variation (natural process variation)
- Assignable cause variation (specific abnormal factors)

A process is considered statistically in control when all observed values fall within established control limits.

- The sample size per batch is constant (200 pipes)
- The data represent number of defective units

An NP-control chart is appropriate for monitoring defectives.

Calculation of Control Limits:

Total defectives = 344
Number of batches = 20

Average number of defectives ($n\bar{p}$):

$n\bar{p}$ = 344 / 20 = 17.2

Proportion defective ($\bar{p}$):

$\bar{p}$ = 17.2 / 200 = 0.086

Control Limits:

UCL = 29.08
LCL = 5.32

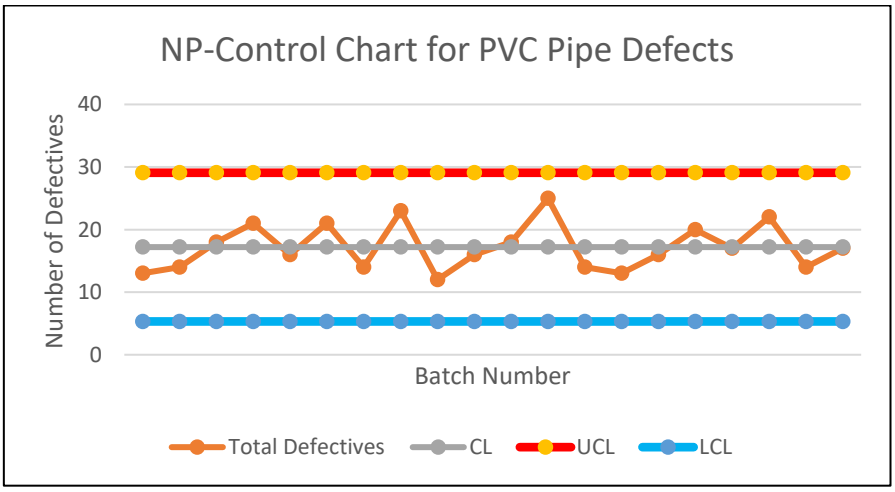

Figure 4: PVC Pipe NP-Control Chart

All observed defect values fall within the calculated control limits. No points exceed the upper or lower control limits, indicating that the production process is statistically stable. The variation observed across batches is attributed to common causes rather than specific abnormal disturbances.

However, although the process is stable, the average defect rate of 8.6% suggests that improvement beyond stability is desirable.

4. Trend Analysis

Trend analysis is conducted to observe whether defect levels show systematic increase or decrease over time. Even if a process is statistically in control, a rising trend may indicate emerging problems.

A line graph was plotted showing batch number versus total defectives.

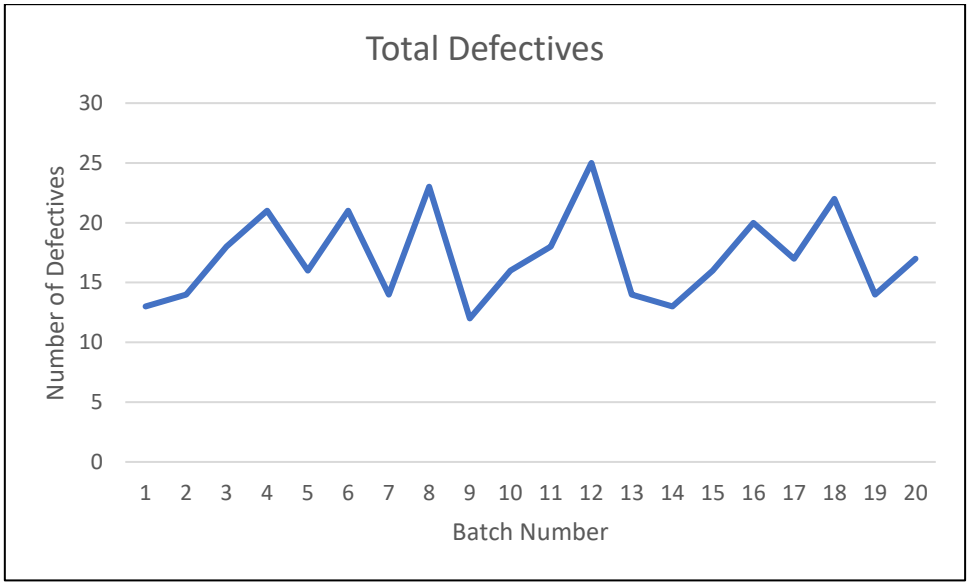

Figure 5: PVC Pipe Defectives Trend Chart

The trend graph shows normal fluctuation in defect levels between 12 and 25 defectives per batch. No consistent upward or downward trend is observed, confirming stable production behaviour during the study period.

5. Cost Impact Analysis

Quality defects not only affect product performance but also result in financial loss. Estimating the economic impact helps management understand the importance of defect reduction.

Assuming a cost of ₹150 per defective pipe:

Total defectives = 344

Total estimated loss = ₹51,600

Defect-wise cost impact:

- Crack = ₹25,800
- Melt = ₹17,700
- Hollow = ₹8,100

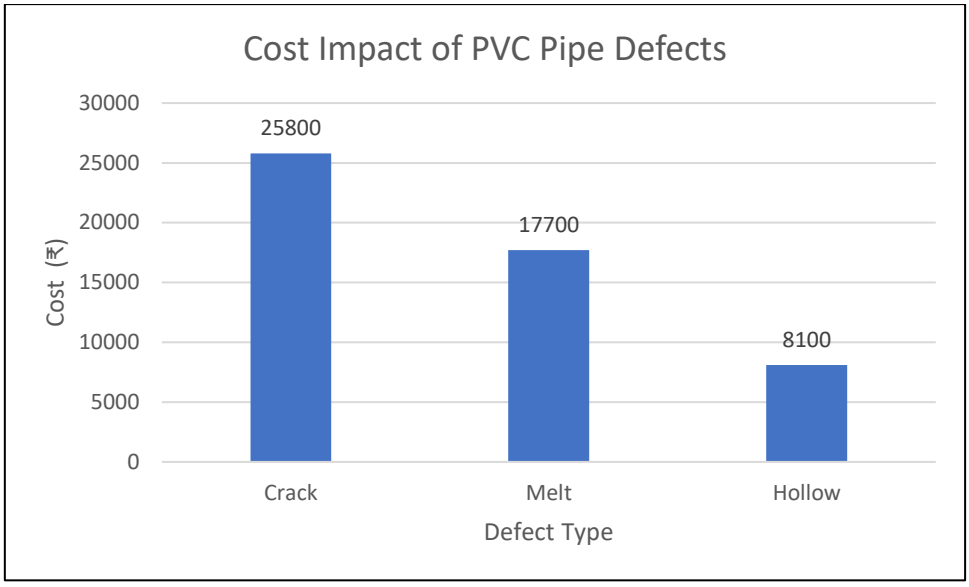

Figure 6: PVC pipe cost impact Chart

Crack defects contribute the highest economic loss, reinforcing the need to prioritize corrective measures for this defect category. Reducing crack defects would provide the greatest financial benefit.

<u>Preventive Quality Monitoring Framework:</u>

To ensure sustained quality performance in PVC pipe manufacturing, a preventive quality monitoring framework is proposed based on Statistical Quality Control principles. The purpose of this framework is to enable early detection of process deviations, reduce defect rates, and maintain continuous process stability.

1. Continuous Monitoring Using Control Charts

An NP-control chart should be implemented at the production floor to monitor the number of defective pipes per batch. Since the sample size remains constant, the NP-chart is suitable for real-time monitoring.

- The center line (CL) represents the average number of defectives.
- The upper and lower control limits define acceptable process variation.
- Any point exceeding the control limits should trigger immediate investigation.

Control charts should be updated after each production batch to detect abnormal variation at an early stage.

2. Out-of-Control Action Plan

If a data point exceeds the upper control limit or shows non-random patterns, the following corrective steps should be taken:

1. Stop production temporarily for inspection.
2. Check machine parameters such as extrusion temperature, pressure, and feeder speed.
3. Inspect raw material consistency and moisture content.
4. Verify operator settings and handling procedures.
5. Record the cause and corrective action taken.

This structured response prevents recurrence of defects and supports systematic problem resolution.

3. Defect Prioritization Strategy

Based on Pareto analysis, crack defects contribute the highest proportion of total defects. Therefore:

- Machine calibration should focus on temperature uniformity.
- Regular maintenance schedules should be enforced.
- Operators should receive training on detecting early signs of cracking.

By prioritizing the most significant defect category, quality improvement efforts become more effective and resource-efficient.

4. Periodic Review and Reporting

Weekly and monthly quality reports should be prepared including:

- Average defect rate
- Control chart status
- Defect category trends
- Cost impact analysis

Management review meetings should be conducted to evaluate process performance and approve preventive measures where necessary.

5. Continuous Improvement Integration

The monitoring system should not only detect problems but also support long-term quality enhancement. Data collected from control charts should be analyzed periodically to:

- Identify recurring patterns
- Optimize machine settings
- Improve process standardization
- Reduce overall defect rate over time

This aligns with the philosophy of continuous improvement in manufacturing systems.

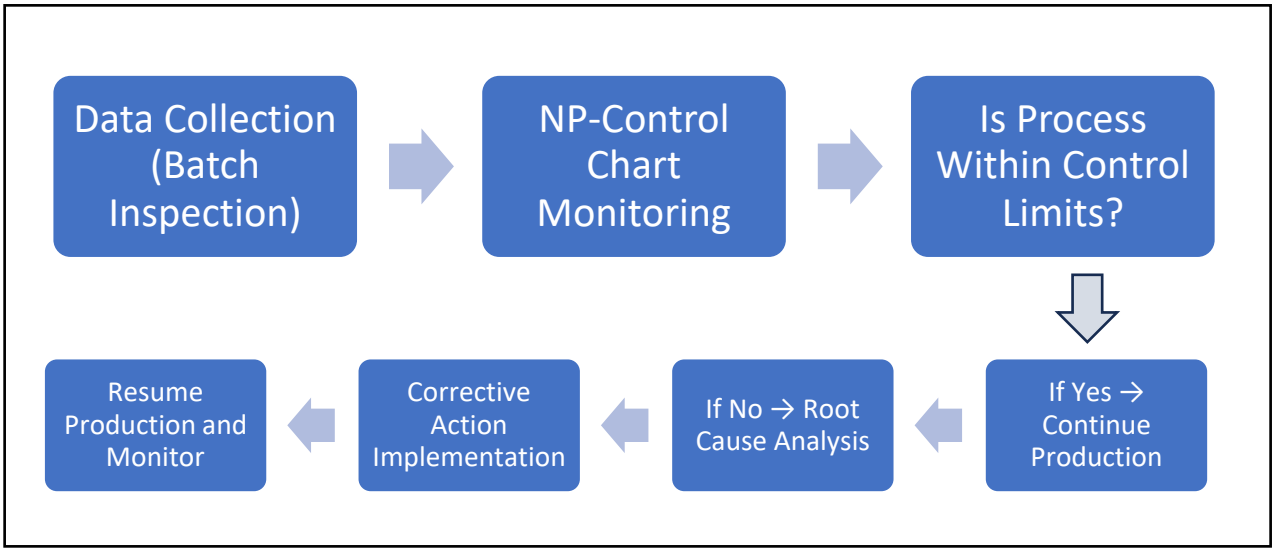

Figure 7: PVC pipe Quality Monitoring Framework

The proposed preventive quality monitoring framework integrates control charts, defect prioritization, structured corrective action, and periodic review. By implementing this system, PVC pipe manufacturers can move from reactive defect correction to proactive process control, ensuring stable production and reduced economic loss.

<u>Findings:</u>

Based on the statistical analysis of the PVC pipe production data, the following findings were observed:

1. Crack defects represent the highest proportion of total defects (50%), making it the most critical quality issue in the production process.
2. The average number of defective pipes per batch is 17.2 out of 200 inspected pipes, indicating an average defect rate of approximately 8.6%.

3. The NP-control chart analysis shows that all sample points lie within the calculated control limits, suggesting that the process is statistically stable and operating under common cause variation.
4. No consistent increasing or decreasing trend was observed in batch-wise defect levels, indicating stable production performance over the observed period.
5. The estimated total economic loss due to defective production amounts to ₹51,600, with crack defects contributing the highest financial impact.

These findings indicate that although the process is statistically in control, there remains significant scope for quality improvement.

Recommendations:

1. Priority should be given to reducing crack defects through improved control of extrusion temperature, cooling rate, and machine alignment.
2. Preventive maintenance and periodic machine calibration should be implemented to minimize process variation.
3. NP-control charts should be used continuously for batch-wise monitoring to detect deviations at an early stage.
4. Operators should be trained on standard operating procedures to ensure consistent production settings.
5. Management should establish a defect reduction target to improve overall process performance beyond current stability levels.

Conclusion:

This study analyzed PVC pipe manufacturing defects using Statistical Quality Control techniques. A total of 4,000 pipes across 20 production batches were evaluated, with 344 defective units identified, resulting in an average defect rate of 8.6%. Pareto analysis revealed that crack defects account for 50% of total defects, making it the most significant quality concern.

The NP-control chart indicated that all defect values fall within the calculated control limits (UCL = 29.08, LCL = 5.32), confirming that the process is statistically stable. However, the estimated economic loss of ₹51,600 highlights the need for further defect reduction efforts.

The proposed preventive quality monitoring framework using control charts supports continuous process monitoring and provides a structured approach for long-term quality improvement in PVC pipe production. Although the process is statistically stable, the defect level remains economically significant and requires reduction efforts.

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